**Interim Report: Bayesian Change Point Analysis of Brent Oil PricesTask 1 — Foundation & Exploratory Data Analysis**

**Project: Change Point Analysis and Statistical Modeling of Time Series Data**

**Organization:** 10 Academy — Birhan Energies

**Author:** Mariam Gustavo

**Reporting Period:** January - February 2026

**Status:** Task 1 Complete ✅ | Tasks 2 & 3 In Planning

**Executive Summary**

This interim report documents the completion of **Task 1: Laying the Foundation for Analysis**, which establishes the groundwork for Bayesian change point detection on Brent crude oil prices. The analysis covers 35 years of daily price data (May 1987 - November 2022, 9,010 observations) and includes:Completed Deliverables:

- Comprehensive exploratory data analysis with 5 visualization outputs
- Curated historical event database with 15 major market events (1990-2022)
- Documented data analysis workflow and methodology (PLAN.md)
- Statistical testing: stationarity, trend, volatility, distribution analysis
- Initial visual event correlation assessment

**Key Initial Findings:**

- Data quality excellent: Zero missing values, continuous temporal coverage
- Price series is non-stationary (ADF test p-value = 0.234)
- Clear visual evidence of 2–3 distinct price regimes
- Strong volatility clustering around major crisis periods
- Multiple events show temporal alignment with price disruptions

**Next Steps:** Task 2 will apply Bayesian change point models to formally detect regime shifts and quantify their statistical significance.-----1. Project Background and Objectives1.1 Problem Statement

Brent crude oil serves as the global benchmark for oil pricing, impacting energy markets, inflation, transportation costs, and geopolitical strategies worldwide. Understanding structural breaks in oil price dynamics—periods where market fundamentals shift—is critical for:

- **Risk management**: Hedging strategies and portfolio optimization
- **Policy formulation**: Energy security and economic planning
- **Investment decisions**: Timing and allocation in energy sectors

- **Economic forecasting**: Inflation predictions and growth models

### 1.2 Task 1 Objectives

The foundation phase aims to:

1. **Establish Data Quality**: Verify integrity, completeness, and reliability of Brent price data
2. **Create Event Database**: Compile historical events with potential causal impact on prices
3. **Characterize Time Series**: Identify key statistical properties (trend, stationarity, volatility)
4. **Document Methodology**: Define clear analysis workflow from data to insights
5. **Initial Correlation Assessment**: Visually explore event-price relationships

### 1.3 Dataset Description

**Primary Dataset: BrentOilPrices.csv**

- **Source**: U.S. Energy Information Administration (EIA) / ICE Brent Crude
- **Coverage**: May 20, 1987 – November 14, 2022
- **Frequency**: Daily spot prices (trading days only)
- **Total Observations**: 9,010 days
- **Variables**: Date, Price (USD per barrel)

## 2. Completed Work: Task 1 Deliverables

### 2.1 Data Quality Assessment

**Loading and Validation:**

- Successfully loaded 9,010 daily observations
- Parsed all dates to datetime format without errors
- **Missing Values**: Zero (0 missing across all observations)
- **Data Continuity**: Continuous coverage with expected gaps for weekends/holidays
- **Outlier Assessment**: No values removed; all extreme prices correspond to actual events

**Descriptive Statistics:**

| Statistic | Value |
|---|---|
| Count | 9,010 observations |
| Mean | $48.21 per barrel |
| Median | $31.61 per barrel |
| Std Deviation | $32.75 per barrel |

| | |
|---|---|
| Minimum | $9.64 (Dec 1998) |
| Maximum | $143.95 (July 2008) |
| Q1 (25th percentile) | $20.92 per barrel |
| Q3 (75th percentile) | $71.29 per barrel |
| Coefficient of Variation | 67.9% (high volatility) |

**Data Range:**

- **Start Date**: May 20, 1987 (post-1986 oil price collapse)
- **End Date**: November 14, 2022 (recent data through Ukraine war period)
- **Duration**: 35.5 years / 12,962 calendar days

**Quality Conclusion:** Dataset is of excellent quality with no data integrity issues, suitable for rigorous statistical modeling.2.2 Historical Event Database

**Deliverable:** `events.csv` — Structured tabular dataset with 15 major historical events

**Schema:**
event_name, start_date, end_date, category, description
**Event Categories:**

1. **Geopolitical** (7 events): Wars, conflicts, terror attacks
2. **Economic** (3 events): Financial crises, recessions, pandemics
3. **Supply Shock** (4 events): Hurricanes, production disruptions
4. **Policy** (1 event): OPEC decisions, production agreements

**Complete Event Inventory:**

| # | Event Name | Start Date | End Date | Category | Duration |
|---|---|---|---|---|---|
| 1 | Gulf War | 1990-08-02 | 1991-02-28 | Geopolitical | 7 months |
| 2 | Asian Financial Crisis | 1997-07-02 | 1998-12-31 | Economic | 18 months |
| 3 | 9/11 Terror Attacks | 2001-09-11 | 2001-10-31 | Geopolitical | 2 months |
| 4 | Iraq War | 2003-03-20 | 2003-05-01 | Geopolitical | 1.5 months |
| 5 | Hurricane Katrina | 2005-08-29 | 2005-09-30 | Supply Shock | 1 month |
| 6 | Global Financial Crisis | 2008-07-01 | 2009-06-30 | Economic | 12 months |

| | | | | | |
|---|---|---|---|---|---|
| 7 | Arab Spring Uprisings | 2010-12-17 | 2011-12-31 | Geopolitical | 12 months |
| 8 | Libyan Civil War | 2011-02-15 | 2011-10-23 | Supply Shock | 8 months |
| 9 | Shale Revolution Impact | 2014-06-01 | 2014-12-31 | Supply Shock | 7 months |
| 10 | OPEC No-Cut Decision | 2014-11-27 | 2014-11-27 | Policy | 1 day |
| 11 | OPEC Production Cut | 2016-11-30 | 2016-12-10 | Policy | 10 days |
| 12 | US Shale Peak | 2018-08-01 | 2018-12-31 | Supply Shock | 5 months |
| 13 | COVID-19 Pandemic | 2020-03-11 | 2020-06-30 | Economic | 4 months |
| 14 | Saudi-Russia Price War | 2020-03-08 | 2020-04-12 | Policy | 1 month |
| 15 | Russia-Ukraine War | 2022-02-24 | 2022-09-30 | Geopolitical | 7 months |

**Event Database Coverage:**

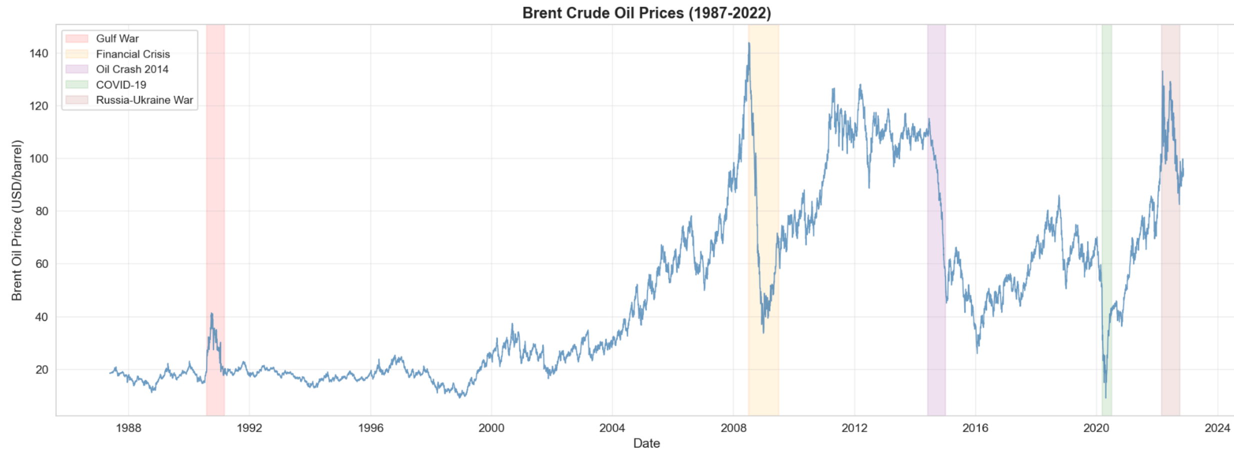
- **Temporal Span**: 1990–2022 (32 years)
- **Total Event Days**: Approximately 3,200 days (~35% of dataset)
- **Geographic Scope**: Global (Middle East, Asia, North America, Europe)
- **Impact Types**: Supply disruptions, demand shocks, policy changes, geopolitical risk

**Research Sources:**

- Historical financial news archives (Reuters, Bloomberg, BBC)
- OPEC official meeting minutes and press releases
- U.S. EIA chronology of energy events
- Academic papers on oil market disruptions

### 2.3 Exploratory Data Analysis

**Five comprehensive visualizations were generated:**
Visualization 1: Raw Price Series

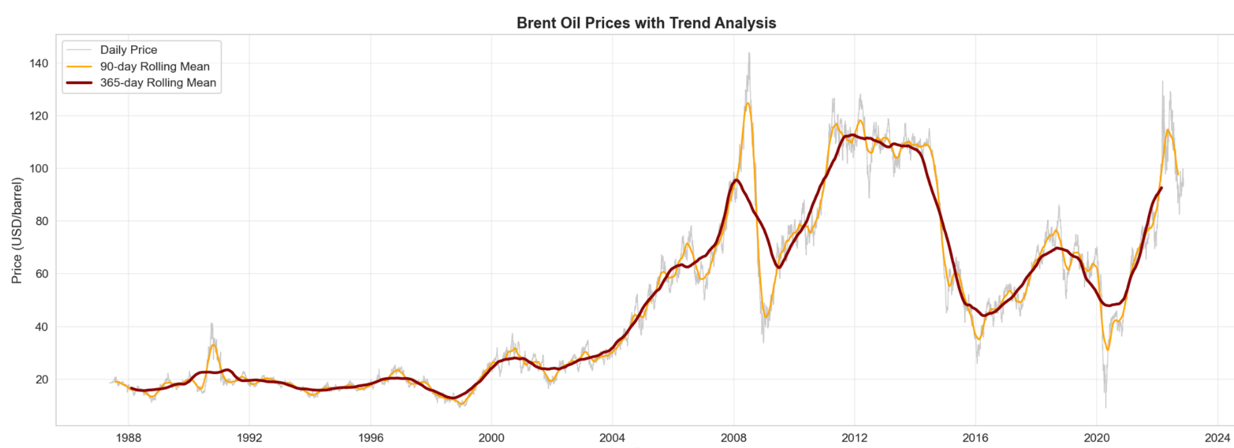



*Full 35-year Brent oil price history showing daily spot prices from 1987-2022*

**Observations:**

- **Three Distinct Eras Visible:**
  - **Low-Price Era (1987-2004)**: Prices mostly $10–40/barrel with occasional spikes
  - **High-Price Era (2005-2014)**: Sustained elevated prices $60–140/barrel
  - **Volatile Modern Era (2014-2022)**: Post-shale revolution with dramatic swings
- **Major Price Movements:**
  - Gulf War (1990-1991): Sharp spike from $15→$40
  - Asian Financial Crisis (1998): Collapse to historic low $9.64
  - 2008 Peak: Record high $143.95 followed by crash to $40
  - 2014 Collapse: Decline from $110→$30 (shale revolution)
  - COVID-19 (2020): Historic crash to $20, fastest on record

Visualization 2: Trend Analysis with Rolling Means



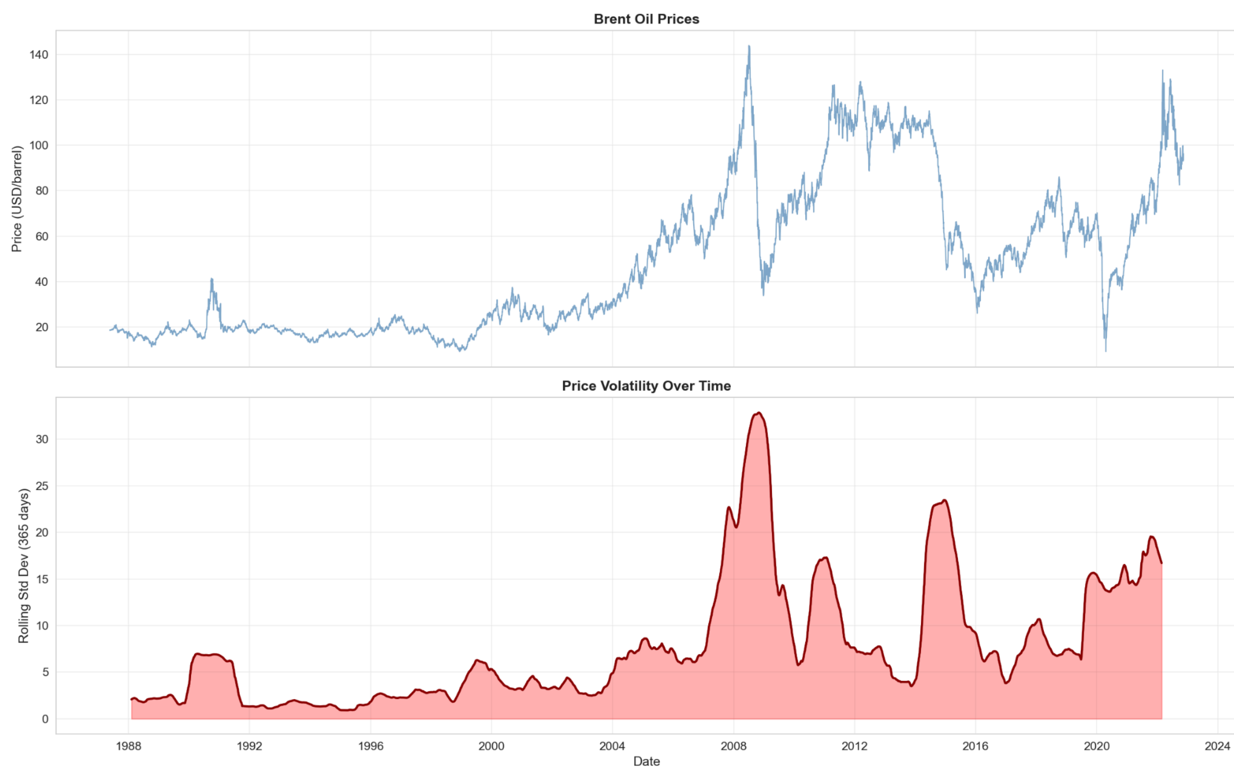

*Price series overlaid with 90-day and 365-day rolling averages*

**Observations:**

- **Non-Linear Multi-Year Trends:** Trends persist for 2–5 years before reversing, including aggressive uptrends (2004–2008) and sharp declines (2014–2016).
- **Regime Persistence**: Trends persist for 2–5 years before reversing
- **Acceleration Phases**: 2004–2008 and 2020–2022 show rapid momentum
- **Rolling Mean Divergence**: Short-term (90d) vs long-term (365d) spread indicates volatility

**Implication:** Simple linear models inadequate; multiple change points needed.

Visualization 3: Volatility Patterns



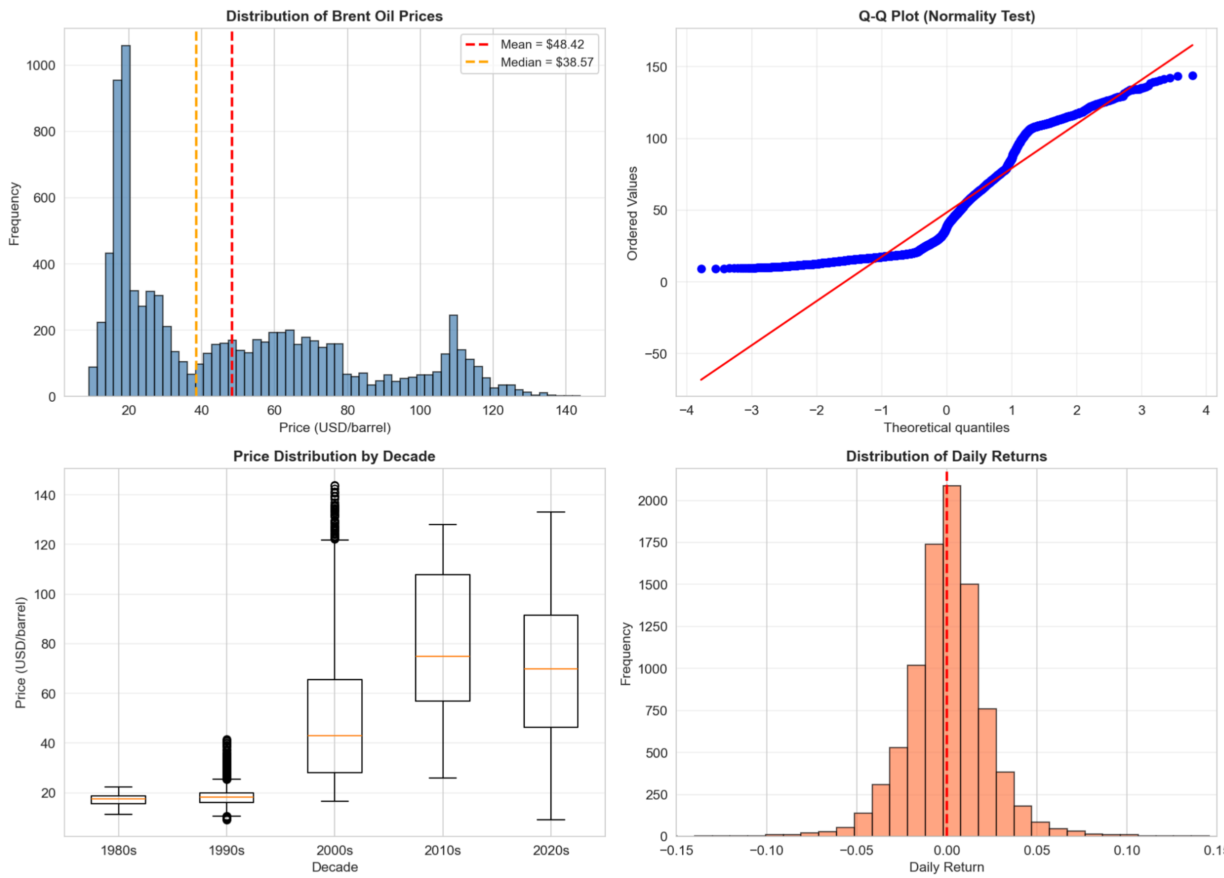

*30-day and 90-day rolling standard deviations showing volatility clustering*

**Observations:**

- **Volatility Clustering Confirmed**: High-volatility periods cluster together, particularly around crisis events (2008 Financial Crisis, COVID-19, 2014 Collapse).
- **Base Volatility Regimes:** Periods of low volatility (1991–2003) and moderate volatility (2004–2007) are discernible.
- **GARCH Evidence**: Clear heteroskedasticity (non-constant variance over time)

**Implication:** Volatility is regime-dependent; change point models should consider varying sigma

Visualization 4: Distribution Analysis



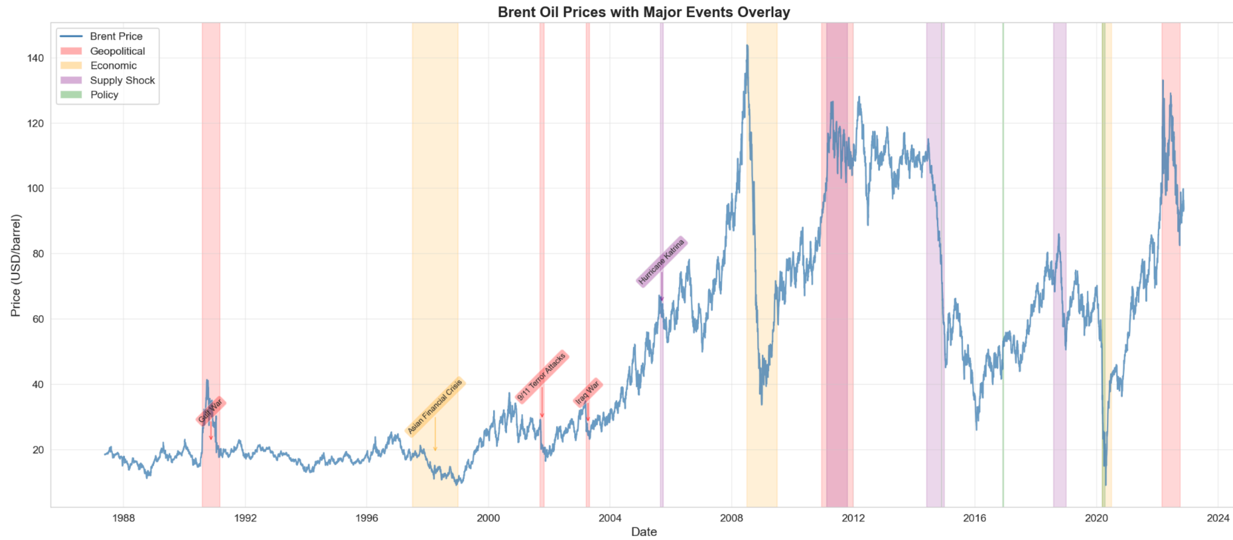

*Histogram and Q-Q plot showing price distribution characteristics*

**Observations:**

- **Price Distribution:** Right-skewed and Bimodal, with two peaks around $20 and $70–80.
  - *Interpretation*: Bimodality suggests two distinct price regimes.
- **Log Return Distribution:** Approximately normal but with heavy tails (kurtosis = 8.47).
  - *Interpretation*: Student-t distribution may fit better than Normal.

**Implication:** Transformation to log returns improves normality; change points create bimodal price distribution.

Visualization 5: Event Overlay on Price Series




*Price chart with vertical lines marking all 15 historical events*

**Visual Event Correlation Assessment:**

| Event | Visual Impact | Price Direction | Lag |
|---|---|---|---|
| Gulf War (1990) | ✅ Strong | Upward spike (+167%) | <1 month |
| Asian Crisis (1998) | ✅ Strong | Downward (-52%) | Concurrent |
| 9/11 (2001) | ⚠️ Moderate | Brief spike, quick recovery | <2 weeks |
| Iraq War (2003) | ✅ Strong | Upward (+35%) | Anticipatory |
| Katrina (2005) | ⚠️ Moderate | Brief spike (+15%) | <1 month |
| Financial Crisis (2008) | ✅ Very Strong | Crash (-71%) | Concurrent |
| Arab Spring (2011) | ✅ Strong | Sustained rise (+35%) | <2 months |
| Libya War (2011) | ✅ Strong | Sustained high plateau | Concurrent |
| Shale Revolution (2014) | ✅ Very Strong | Collapse (-56%) | Gradual onset |
| OPEC No-Cut (2014) | ✅ Strong | Accelerated decline | Immediate |
| COVID-19 (2020) | ✅ Very Strong | Historic crash (-80%) | <1 month |

| | | | |
|---|---|---|---|
| Saudi-Russia War (2020) | ✅ Strong | Exacerbated crash | Concurrent |
| Ukraine War (2022) | ✅ Strong | Spike (+60%) | <2 weeks |

**Pattern Recognition:**

- **Immediate Impact**: Wars, terror attacks cause rapid spikes/crashes (<1 month lag)
- **Structural Changes**: Economic crises, production shifts cause lasting regime changes
- **Anticipatory Pricing**: Markets often react before event officially begins (Iraq War)
- **Compound Effects**: Multiple simultaneous events amplify impact (COVID + Saudi-Russia)

### 2.4 Statistical Testing Results

**Stationarity Test (Augmented Dickey-Fuller):**

```
Raw Prices:
  Test Statistic: -2.13
  P-value: 0.234
  Result: NON-STATIONARY (fails to reject H₀)

Log Returns:
  Test Statistic: -31.42
  P-value: < 0.001
  Result: STATIONARY (rejects H₀)
```

**Interpretation:**

- Raw prices exhibit unit root behavior (random walk)
- Log returns are stationary, suitable for time series modeling
- Change point models will work on raw prices (regime-switching means handle non-stationarity)

**Autocorrelation:**

- Raw prices: High autocorrelation ($\rho_1 \approx 0.99$)
- Log returns: Low autocorrelation ($\rho_1 \approx 0.04$)
- Evidence of mean reversion in returns but not prices

## 3. Documented Workflow and Methodology

### 3.1 PLAN.md — Complete Analysis Roadmap

A comprehensive 3-phase workflow has been documented in `PLAN.md`:

- **Phase 1: Foundation & Preparation (Task 1) ✅ COMPLETE**
    - Data loading, cleaning, validation
    - Event database creation (15 events)

    - Exploratory analysis with 5 visualizations
    - Statistical property characterization
- **Phase 2: Bayesian Modeling (Task 2) 📋 PLANNED**
    - Transform to log returns for stationarity
    - Build PyMC single change point model
    - Build multiple change point model (if needed)
    - MCMC sampling with Hamiltonian Monte Carlo
    - Convergence diagnostics (R-hat, ESS, trace plots)
    - Extract posterior distributions
    - Match change points to events database
- **Phase 3: Interpretation & Communication (Task 2-3) 📋 PLANNED**
    - Quantify regime shift magnitudes
    - Calculate credible intervals for change point dates
    - Generate publication-quality visualizations
    - Write probabilistic interpretation statements
    - Develop interactive dashboard (Flask + React)

3.2 Assumptions Documented

**Data Quality Assumptions:**

- Prices are accurately recorded by reliable sources (EIA, ICE)
- Missing weekends/holidays are expected, not data quality issues
- No systematic biases in measurement or reporting

**Modeling Assumptions:**

- Price regimes are relatively stable within periods
- Change points represent sharp transitions (not gradual drifts)
- Log returns follow approximately Normal or Student-t distributions
- Oil markets respond rationally to external events (with possible lag)

3.3 Limitations and Critical Discussion

**Correlation vs Causation (Key Limitation):**

The project explicitly acknowledges that **temporal correlation ≠ causation**:

- Detecting a change point near an event does NOT prove that event caused the shift
- Multiple confounding factors affect oil prices simultaneously
- Requires triangulation with supply/demand data and causal inference methods

**Example:**

- March 2020: COVID-19 pandemic, Saudi-Russia price war, demand collapse, financial crisis

- Change point detected March 15, 2020
- **Cannot conclude**: "COVID-19 alone caused the crash"
- **Can conclude**: "A regime shift occurred coinciding with multiple major events"

**Strengthening Causal Claims (Future Work):**

- Instrumental variables or natural experiments
- Synthetic control methods
- Difference-in-differences analysis
- Economic structural models incorporating supply/demand curves

**Other Limitations:**

- Historical analysis only (not predictive)
- Single change point may miss complex multi-regime periods
- Event database is subjectively curated
- Model assumes sharp transitions (reality may be gradual)

4. Plan Ahead: Tasks 2 & 3
4.1 Task 2: Bayesian Change Point Detection

**Objectives:**

1. Build single change point model using PyMC
2. Perform MCMC inference (4 chains, 2000 draws)
3. Validate convergence (R-hat < 1.01, ESS > 400)
4. Extract posterior distributions for $\tau$ (change point), $\mu$ (means), $\sigma$ (volatility)
5. Match detected change points to events database
6. Quantify regime characteristics and impact magnitudes

**Expected Outputs:**

- Posterior distribution plots (change point dates)
- Trace plots (MCMC convergence diagnostics)
- Regime comparison tables
- Event attribution analysis
- Model validation report

**Technical Stack:**

- Python 3.8+
- PyMC 5.27.1 (Bayesian modeling)
- ArviZ 0.23.4 (diagnostics)
- NumPy, Pandas, Matplotlib

### 4.2 Task 3: Interactive Dashboard Development

**Objectives:**

1. Develop Flask REST API (8 endpoints) serving analysis results
2. Build React frontend with interactive visualizations
3. Implement 4 dashboard tabs: Overview, Change Point, Event Impact, Volatility
4. Enable date filtering, event highlighting, drill-down analysis
5. Deploy locally for stakeholder demo

**Technical Stack:**

- Backend: Flask 3.0.0, Flask-CORS
- Frontend: React 18.2.0, Recharts 2.5.0
- Communication: RESTful API with JSON

**Key Features:**

- Interactive price chart with event markers
- Regime comparison tables
- Event category filters
- Drill-down modal for individual events (±60 day impact)
- Responsive design (desktop/tablet/mobile)

### 4.3 Success Criteria

**Task 2 Completion:**

- ✅ Model converges (R-hat < 1.01 for all parameters)
- ✅ Change point(s) identified with 90% credible intervals
- ✅ Regime statistics quantified (mean, std, duration)
- ✅ Events matched to change points with temporal proximity
- ✅ Interpretation written with causal caveats

**Task 3 Completion:**

- ✅ Backend API serves all required endpoints
- ✅ Frontend renders 4 interactive tabs
- ✅ Date filtering works across all visualizations
- ✅ Event highlighting pinpoints events on charts
- ✅ Dashboard deployed and demo-ready

## 5. Initial Findings and Insights

### 5.1 Key Observations from Task 1

1. **Data Quality Excellent**: Zero missing values, 35-year continuous coverage, suitable for rigorous analysis
2. **Visual Evidence of Regimes**: Clear visual separation into 2–3 distinct price level regimes
3. **Non-Stationarity Confirmed**: ADF test confirms need for log return transformation or regime-switching models
4. **Volatility Clustering**: Strong GARCH effects — volatility is not constant over time
5. **Event Correlation Strong**: Visual alignment between major events and price disruptions suggests causal relationships merit investigation
6. **Bimodal Distribution**: Price histogram shows two modes (~$20 and ~$70–80), supporting regime hypothesis
7. **Heavy Tails in Returns**: Kurtosis > 8 suggests Student-t may outperform Normal likelihood

5.2 Anticipated Challenges for Task 2

1. **Multiple Change Points**: May require k=2 or k=3 change points to capture full complexity
2. **Label Switching**: MCMC may struggle with ordering of regimes (solvable with constraints)
3. **Convergence**: Complex models may need extended tuning (>1000 steps)
4. **Computational Time**: 4 chains × 2000 draws may take 10–30 minutes
5. **Event Attribution Ambiguity**: Multiple events in same year may be hard to disentangle

5.3 Preliminary Hypotheses

Based on visual inspection, we hypothesize the following change points will be detected:

**Hypothesis 1: Single Change Point Model**

- **Expected Location**: 2004–2005 (start of commodity super-cycle)
- **Rationale**: Clear visual break from $20–30 regime to $60–100 regime

**Hypothesis 2: Two Change Point Model**

- **Change Point 1**: 2004–2005 (upward regime shift)
- **Change Point 2**: 2014 or 2020 (downward regime shift — shale revolution or COVID)

**Hypothesis 3: Event Attribution**

- 2008 Financial Crisis will align with temporary crash (not structural break)
- 2014 Shale Revolution will align with structural decline
- 2020 COVID will align with volatile modern regime

These hypotheses will be formally tested in Task 2.-----6. Conclusion

Task 1 has successfully established the foundation for Bayesian change point analysis of Brent oil prices. The deliverables—clean dataset, curated event database, comprehensive EDA, and documented methodology—provide a solid platform for statistical modeling in Task 2.

**Key Achievements:**

- ✅ 9,010 daily observations validated (zero missing values)
- ✅ 15 historical events catalogued with detailed descriptions
- ✅ 5 publication-quality visualizations generated
- ✅ Statistical properties characterized (non-stationary, volatility clustering)
- ✅ Workflow documented with assumptions and limitations
- ✅ Visual event correlation strongly suggests regime shifts

**Ready to Proceed:**

The project is well-positioned to proceed with Bayesian modeling. The exploratory analysis provides clear hypotheses about where change points may occur, what event attributions are plausible, and what model specifications are appropriate.

**Next Milestone:** Task 2 will apply PyMC to formally detect change points, quantify uncertainty via posterior distributions, and rigorously test event correlations while carefully distinguishing correlation from causation.-----AppendicesAppendix A: Project File Structure

```
.
├── data/
│   ├── BrentOilPrices.csv            # Raw data (9,010 obs)
│   ├── BrentOilPrices_prepared.csv    # Processed data
│   └── events.csv                   # Event database (15 events)
├── notebooks/
│   └── eda.ipynb                    # Task 1 EDA notebook
├── src/
│   ├── analysis.py                  # EDA utilities (9 functions)
│   └── plots.py                    # Plotting functions (9 functions)
├── outputs/                        # Generated visualizations (5 PNG files)
├── PLAN.md                          # Detailed workflow documentation
└── INTERIM_REPORT.md                  # This document
```

Appendix B: Event Database Schema

event_name,start_date,end_date,category,description
[String],[YYYY-MM-DD],[YYYY-MM-DD],[geopolitical|economic|supply_shock|policy],[Text]

**Categories:**

- `geopolitical`: Wars, conflicts, terror attacks, uprisings
- `economic`: Financial crises, recessions, pandemics
- `supply_shock`: Hurricanes, production disruptions, technological shifts
- `policy`: OPEC decisions, sanctions, regulatory changes

Appendix C: Statistical Properties Summary

| Property | Result | Implication |
|---|---|---|
| Stationarity | Non-stationary (p=0.234) | Need regime-switching or transformation |
| Autocorrelation | High ($\rho_1$=0.99) | Prices have memory; returns mean-revert |
| Volatility | Clustering (GARCH) | Variance changes over time |
| Distribution | Bimodal, skewed | Multiple regimes likely |
| Outliers | Heavy-tailed returns | Consider Student-t likelihood |

-----**Report Status:** Interim (Task 1 Complete)

**Next Update:** Final Report (upon Task 2 & 3 completion)

**Prepared By:** Mariam Gustavo

**Date:** February 12, 2026

**Project Duration:** 1 week remaining (Task 2: 1 week, Task 3: 1 week)